MATH 2101 LINEAR ALGEBRA I, FALL 2025 – ASSIGNMENT 7 OUTLINE OF SOLUTION

Due date: 20th November (Thursday), 2025 10:00pm.

- To receive *full credits*, the solution has to be *clear* and provide sufficient explanations.
- All solutions have to be turned in HKU moodle in the format of *PDF file*.
- Please include your *Name, UID, Faculty, Major (if declared)* in your solution.

1. Determine if each of the following statements is true: Let A be a 3×3 matrix. (*We assume that all roots in its characteristic polynomial are real.*)
   (a) If A is diagonalizable, then A^2 is also diagonalizable.
   (b) If A^2 is diagonalizable and A is invertible, then A is also diagonalizable.
   (c) If A^3 has 3 distinct eigenvalues, then A is diagonalizable.

   If the statement is true, provide a proof. If the statement is false, provide a counterexample.

Sol.

(a) True. Since A is diagonalizable, one can find an invertible 3×3 matrix P such that $P^{-1}AP = D$ for some diagonal matrix D. Then

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD^2P^{-1},$$

where D^2 is clearly diagonal. In other words, A^2 is diagonalizable.

(b) True. This problem requires some results from MATH2102. Take a look at Conard's note on minimal polynomials. I give a sketch of proof using Theorem 4.11 of the note, which asserts the following: A linear transformation $T : V \to V$ is diagonalizable (in $\mathbb{R}$) if and only if its minimal polynomial in $\mathbb{R}[x]$ splits in $\mathbb{R}[x]$ and has distinct roots. Consider the minimal polynomial $m_2(x)$ of A^2, which must be in the form

$$m_2(x) = \prod_{i=1}^{k} (x - a_i), \quad a_i \in \mathbb{R} \setminus \{0\},$$

for some $k \in \{1, 2, 3\}$. Here, a_i are distinct by the given theorem. Moreover, since a_i are eigenvalues of A^2, they have to be the squares of eigenvalues of A, and hence positive. By the definition of minimal polynomial, we have

$$M_3(\mathbb{R}) \ni 0 = m_2(A^2) = \prod_{i=1}^{k} (A^2 - a_i I_3) = \prod_{i=1}^{k} (A - \sqrt{a_i} I_3)(A + \sqrt{a_i} I_3).$$

As a consequence, the minimal polynomial $m_1(x)$ of A satisfies that $m_1(x) \mid f(x)$, where $f(x) = \prod_{i=1}^{k}(x - \sqrt{a_i})(x + \sqrt{a_i})$ as one observes that $f(A) = 0$ (together with Theorem 4.4 of Conard's note). Then $m_1(x)$ has to be a polynomial which splits in $\mathbb{R}[x]$ and has distinct roots. Whence A is diagonalizable.

Remark. The argument above is still a little hand-wavy for the course (even after adding Conrad's note, where he uses field extension (e.g., $\mathbb{R} \hookrightarrow \mathbb{C}$) in the proof of Theorem 4.7, and the notion of direct sum), as we need more statements and results in the ring of polynomials over the real numbers, and the notion of divisibility. You will have a systematic study of minimal polynomials and the Cayley-Hamilton theorem in MATH2102, as well as the notions of divisibility and polynomial rings in MATH3304, MATH3301, and MATH4302. In the meantime, one can consider applying the Jordan normal form theorem to overkill the problem.

(c) True. If A^3 has 3 distinct eigenvalues μ_1, μ_2, μ_3, then A has 3 distinct eigenvalues $\sqrt[3]{\mu_1}$, $\sqrt[3]{\mu_2}$, $\sqrt[3]{\mu_3}$. Whence A is diagonalizable.

2. Determine all the possible values of $x \in \mathbb{R}$ such that the following matrix
$$\begin{pmatrix} 2+x & x \\ -x & -x+2 \end{pmatrix}$$
is diagonalizable.

Sol. We claim that $x = 0$ is the only possible value such that the matrix $A := \begin{pmatrix} 2+x & x \\ -x & -x+2 \end{pmatrix}$ is diagonalizable. We first compute the characteristic polynomial $p(t)$ of A:
$$p(t) = \det(A - tI_2) = \det \begin{pmatrix} 2+x-t & x \\ -x & -x+2-t \end{pmatrix} = (2-t)^2 - x^2 + x^2 = (2-t)^2.$$
It follows that 2 is the only eigenvalue of A. Note that A is diagonalizable if and only if its geometric multiplicity equals its algebraic multiplicity, i.e., $\dim(N(A - 2I_2)) = 2$. Since $A - 2I_2 = \begin{pmatrix} x & x \\ -x & -x \end{pmatrix}$, its null space is $\mathbb{R}^2$ if $x = 0$ and $\operatorname{span}_{\mathbb{R}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ if $x \neq 0$. Therefore, A is diagonalizable if and only if $x = 0$.